

Summary 5

→ all linear combinations of those elements.

① The following generated subspaces:

a) $\langle 1, x, x^2 \rangle$ in $\mathbb{R}[x]$ - real vector space → we need a linear combination

b) $\langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \rangle$ in $M_2(\mathbb{R})$

$$V = A \oplus B \text{ if } \begin{cases} V = A + B \\ A \cap B = \{0\} \end{cases}$$

if $(\forall) v \in V, \exists! s \in S, t \in T \text{ st. } v = s + t.$

$f: A \rightarrow B$ endomorphism if $\begin{cases} A = B \\ \text{homomorphism} \end{cases}$

$$\text{Ker}(f) = \{x \in \mathbb{R} \mid f(x) = 0\}$$

$$\text{Im}(f) = \{f(x) \mid x \in \mathbb{R}\}$$

→ we write the linear combination of these 3 polynomials.

$$a) \langle 1, x, x^2 \rangle = \{a + bx + cx^2 \mid a, b, c \in \mathbb{R}\} = \mathbb{R}_2[x].$$

$$b) \langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \rangle =$$

$$= \left\{ a \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \cdot \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \cdot \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \cdot \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mid a, b, c, d \in \mathbb{R} \right\}$$

$$= \left\{ \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix} \right\}$$

$$= \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{R} \right\}$$

$= M_2(\mathbb{R})$ a) → we obtain all the polynomials with the degree = 2.

b) → " — " all the matrixes 2×2 .

② The following subspaces of $\mathbb{R}^3$ - real vector space.

a) $A = \{(x, y, z) \in \mathbb{R}^3 \mid x=0\} \rightarrow \text{plane } yz \rightarrow \text{origin}$

$(0, y, z)$ we can write it as $y(0, 1, 0) + z(0, 0, 1)$

$\Rightarrow A = \langle (0, 1, 0), (0, 0, 1) \rangle$

$\Rightarrow$ the min. nr. of generators is 2. Can we "generate the origin"?

$a(0, 1, 0) + b(0, 0, 1) = (0, 0, 0) \Leftrightarrow a=b=0.$

b) $B = \{(x, y, z) \in \mathbb{R}^3 \mid x+y+z=0\}$

$\hookrightarrow x = -y-z$

$(x, y, z) = (-y-z, y, z) = y(-1, 1, 0) + z(-1, 0, 1)$

$\Rightarrow B = \langle (-1, 1, 0), (-1, 0, 1) \rangle$

$a(-1, 1, 0) + b(-1, 0, 1) = (0, 0, 0)$

$\Rightarrow (-a, a, 0) + (-b, 0, b) = (0, 0, 0)$

$\Rightarrow (-a-b, a, b) = (0, 0, 0)$

$\Rightarrow a=b=0.$

c) $C = \{(x, y, z) \in \mathbb{R}^3 \mid x=y=z\}$

$(x, y, z) = (x, x, x) = x(1, 1, 1)$

$C = \langle (1, 1, 1) \rangle.$

$\Rightarrow$ a line through the origin

Template: 1) parametrizăm

2) separăm scalarii/coef. la vectorii care apar

3) un set independent (2 $\rightarrow$ plan; 1 $\rightarrow$ dr.)

✓ (3) Following vectors in $\mathbb{R}^3$ real vector space

$$a = (-2, 1, 3)$$

$$b = (3, -2, -1)$$

$$c = (1, -1, 2)$$

$$d = (-5, 3, 4)$$

$$e = (-9, 5, 10)$$

? $\langle a, b \rangle \stackrel{?}{=} \langle c, d, e \rangle$

↳ In order for those to be equal, we may show that, for example, the vectors c, d, e can be written as a linear combination of vectors a, b .

We see that: $c = a + b$

$$(\Rightarrow) (1, -1, 2) = (-2, 1, 3) + (3, -2, -1)$$

$$(\Rightarrow) (1, -1, 2) = (1, -1, 2)$$

$$\underline{d = a - b} \quad (2)$$

$$(\Rightarrow) (-5, 3, 4) = (-2, 1, 3) - (3, -2, -1)$$

$$(\Rightarrow) (-5, 3, 4) = (-5, 3, 4)$$

$$\underline{e = 3a - b} \quad / \quad c + 2d \quad (3)$$

$$(\Rightarrow) (-9, 5, 10) = 3(-2, 1, 3) - (3, -2, -1)$$

$$\begin{aligned} (\Rightarrow) (-9, 5, 10) &= (-6, 3, 9) - (3, -2, -1) \\ &= (-9, 5, 10) \end{aligned}$$

$\Rightarrow c, d, e$ linear comb. of a, b . $\Rightarrow \langle c, d, e \rangle \subseteq \langle a, b \rangle$.

But we also need $\langle a, b \rangle \subseteq \langle c, d, e \rangle$.

from (2) $\Rightarrow a = \underline{b} + d \Rightarrow a = \underline{3a - e} + d$

(3) $\Rightarrow b = 3a - e \Rightarrow -2a = d - e$

$$\Rightarrow a = \frac{1}{2}(e - d)$$

$$b = 3 \cdot \frac{1}{2}(e - d) - e$$

$\Rightarrow a, b$ linear comb. of d, e $\Rightarrow \langle a, b \rangle \subseteq \langle c, d, e \rangle$.

$$\left. \begin{aligned} \langle a, b \rangle &\subseteq \langle c, d, e \rangle \\ \langle c, d, e \rangle &\subseteq \langle a, b \rangle \end{aligned} \right\} \Rightarrow \langle a, b \rangle = \langle c, d, e \rangle$$

$$\textcircled{4} \quad S = \{ (x, y, z) \in \mathbb{R}^3 \mid x+y+z=0 \}$$

$$T = \{ (x, y, z) \in \mathbb{R}^3 \mid x=y=z \}$$

S, T subspaces of $\mathbb{R}^3$ and $\mathbb{R}^3 = S \oplus T$?

$$S \text{ subspace if: } \begin{cases} S \neq \emptyset \\ (\forall) \alpha, \beta \in \mathbb{R}, (\forall) v_1, v_2 \in S, \alpha \cdot v_1 + \beta \cdot v_2 \in S \end{cases}$$

$$(0, 0, 0) \in S \Rightarrow S \neq \emptyset \quad (1)$$

$$\text{Let } (x_1, y_1, z_1), (x_2, y_2, z_2) \in S$$

$$\Rightarrow x_1 + y_1 + z_1 = 0$$

$$\Rightarrow x_2 + y_2 + z_2 = 0.$$

$$\text{Let } \alpha, \beta \in \mathbb{R} : \alpha(x_1, y_1, z_1) + \beta(x_2, y_2, z_2) =$$

$$= (\alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2) \in S$$

$$\Leftrightarrow \alpha x_1 + \beta x_2 + \alpha y_1 + \beta y_2 + \alpha z_1 + \beta z_2 = 0.$$

$$\Leftrightarrow \alpha(\underbrace{x_1 + y_1 + z_1}_0) + \beta(\underbrace{x_2 + y_2 + z_2}_0) = 0 \quad (2)$$

$$x = -y - z \quad (1), (2) \Rightarrow S \text{ subspace of } \mathbb{R}^3.$$

$$(x, y, z) = (-y - z, y, z) = y(-1, 1, 0) + z(-1, 0, 1)$$

$$\Rightarrow S = \langle (-1, 1, 0), (-1, 0, 1) \rangle$$

$$T \text{ subspace if: } \begin{cases} T \neq \emptyset \\ (\forall) \alpha, \beta \in \mathbb{R}, (\forall) v_1, v_2 \in T, \alpha \cdot v_1 + \beta \cdot v_2 \in T \end{cases}$$

$$(0, 0, 0) \in T \Rightarrow T \neq \emptyset \quad (3)$$

$$\alpha(x, x, x) + \beta(x', x', x') = (\alpha x + \beta x', \alpha x + \beta x', \alpha x + \beta x') \in T$$

$$\Leftrightarrow \alpha x + \beta x' = \alpha x + \beta x' = \alpha x + \beta x' \text{ true. } (4)$$

$$(x, x, x) = x(1, 1, 1)$$

$$\Rightarrow T = \langle (1, 1, 1) \rangle$$

$$(3), (4) \Rightarrow T \text{ subspace of } \mathbb{R}^3.$$

$$\mathbb{R}^3 = S \oplus T \text{ if } \begin{cases} \mathbb{R}^3 = S + T & (*) \\ S \cap T = \{0\} & (**) \end{cases}$$

$$\text{if } (v) v \in \mathbb{R}^3, \exists! s \in S \text{ s.t. } v = s + t. \\ \exists! t \in T$$

$$\underline{S+T = \mathbb{R}^3}$$

$$v = (v_1, v_2, v_3) \in \mathbb{R}^3. \Rightarrow (v_1, v_2, v_3) = a(-1, 1, 0) + b(-1, 0, 1) + c(1, 1, 1)$$

$$a, b, c \in \mathbb{R}?$$

$$\underline{s = a(-1, 1, 0) + b(-1, 0, 1) \in S.}$$

$$\underline{t = c(1, 1, 1) \in T}$$

$$(v_1, v_2, v_3) = a(-1, 1, 0) + b(-1, 0, 1) + c(1, 1, 1) \\ = (-a, a, 0) + (-b, 0, b) + (c, c, c)$$

$$\Rightarrow \begin{cases} v_1 = -a - b + c & (1) \\ v_2 = a + c & (2) \\ v_3 = b + c & (3) \end{cases}$$

$$(2) - (3) \Rightarrow v_2 - v_3 = a - b$$

$$v_1 = -a - b + c$$

$$v_2 - v_3 - v_1 = c$$

$$v_2 - v_3 - v_1 = 2a + c$$

$$(2): \cancel{v_2} = a + \cancel{v_2} - \cancel{v_3} - \cancel{v_1} \\ \Rightarrow a = v_1 + v_3.$$

$$\Rightarrow c = v_2 - v_3 - v_1 - 2a.$$

$$(1): \cancel{v_1} = \cancel{v_1} + \cancel{v_3} - b + c$$

$$\Rightarrow$$

$$(2): \cancel{v_2} = a + \cancel{v_2} - \cancel{v_3} - \cancel{v_1} - 2a$$

$$\Rightarrow a = v_3 - v_1$$

$$(1) + (2) \Rightarrow N_1 + N_2 = 2c - b \Rightarrow b = 2c - N_1 - N_2$$

$$(3): N_2 = 2c - N_1 - N_2 + c$$

$$2) 2N_2 = 3c - N_1$$

$$\Rightarrow 3c = 2N_2 + N_1 \Rightarrow c = \frac{1}{3}(2N_2 + N_1)$$

$$b = 2 \cdot \frac{1}{3}(2N_2 + N_1)$$

$$\begin{cases} a = -\frac{1}{3}N_1 + \frac{2}{3}N_2 - \frac{1}{3}N_3 \\ b = -\frac{1}{3}N_1 - \frac{1}{3}N_2 + \frac{2}{3}N_3 \\ c = \frac{1}{3}(N_1 + N_2 + N_3) \end{cases} \quad \text{unique}$$

$$\Rightarrow N = S + T \Rightarrow S + T = \mathbb{R}^3 (*)$$

$$\underline{S \cap T = \{0\}}$$

$$S \cap T = \{u\} \Rightarrow u = (x, y, z) \begin{cases} S \rightarrow x+y+z=0 \\ T \rightarrow x=y=z \end{cases} \Rightarrow u = (0, 0, 0)$$

$$\Rightarrow \text{any writing } n = s + t \begin{cases} s \in S \\ t \in T \end{cases} \text{ is unique and } \mathbb{R}^3 = S \oplus T.$$

$$\checkmark \textcircled{5} \quad S \rightarrow \text{even} \quad \mathbb{R}^{\mathbb{R}} = \{f: \mathbb{R} \rightarrow \mathbb{R}\}$$

$$T \rightarrow \text{odd}$$

a) S, T subspaces of $\mathbb{R}^{\mathbb{R}}$

$$b) \mathbb{R}^{\mathbb{R}} = S \oplus T$$

$$S = \{f \mid f(-x) = f(x)\}$$

$$T = \{f \mid f(-x) = -f(x)\}$$

a) $\bullet$ S subspace if $\begin{cases} S \neq \emptyset \\ (\forall) \alpha, \beta \in \mathbb{R}, (\forall) f, g \in S, \alpha f + \beta g \in S \end{cases}$

$$\theta(x) = 0 = \theta(-x) \in S \Rightarrow S \neq \emptyset \quad (1)$$

$$f, g \in S; \alpha f + \beta g \in S \Leftrightarrow$$

$$(1) (\alpha f + \beta g)(-x) = (\alpha f + \beta g)(x)$$

$$\Leftrightarrow \alpha f(-x) + \beta g(-x) = \alpha f(x) + \beta g(x)$$

$$\Leftrightarrow \alpha f(x) + \beta g(x) = \alpha f(x) + \beta g(x) \text{ true, } (\forall) f, g \in S$$

(1), (2) $\Rightarrow S$ subspace of $\mathbb{R}^{\mathbb{R}}$

b) $\bullet$ T subspace if $\begin{cases} T \neq \emptyset \\ (\forall) \alpha, \beta \in \mathbb{R}, (\forall) f, g \in T, \alpha f + \beta g \in T \end{cases}$

$$\theta(-x) = -\theta(x) = 0 \Rightarrow T \neq \emptyset \quad (1)$$

$$\alpha f + \beta g \in S$$

$$(1) (\alpha f + \beta g)(-x) = -(\alpha f + \beta g)(x)$$

$$(2) \alpha f(-x) + \beta g(-x) = -(\alpha f(x) + \beta g(x))$$

$$\Leftrightarrow \alpha (-f(x)) + \beta (-g(x)) = -(\alpha f(x) + \beta g(x))$$

$$\Leftrightarrow -(\alpha f(x) + \beta g(x)) = -(\alpha f(x) + \beta g(x))$$

$$\Leftrightarrow -(\alpha f(x) + \beta g(x)) = -(\alpha f(x) + \beta g(x)) \text{ true (2)}$$

(1), (2) $\Rightarrow T$ subspace of $\mathbb{R}^{\mathbb{R}}$

$$T \subseteq \mathbb{R}^{\mathbb{R}}$$

$$\checkmark b) \mathbb{R}^{\mathbb{R}} = S \oplus T$$

$$\begin{cases} \mathbb{R}^{\mathbb{R}} = S \oplus T & (1) \\ S \cap T = \{0\} & (2) \end{cases}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$g \in S \Rightarrow g(-x) = g(x)$$

$$h \in T \Rightarrow h(-x) = -h(x)$$

$$f(x) = g(x) + h(x) \Rightarrow h(x) = f(x) - g(x)$$

$$\left(\begin{array}{l} x' = -x \\ \text{replace} \end{array} \right) f(-x) = g(-x) + h(-x) = g(x) - h(x)$$

$$f(-x) = g(x) - h(x)$$

$$\Rightarrow g(x) = f(-x) + h(x)$$

$$= f(-x) + f(x) - g(x)$$

$$\Rightarrow 2g(x) = f(x) + f(-x)$$

$$\Rightarrow g(x) = \frac{1}{2}(f(x) + f(-x)) \in S.$$

$$h(x) = g(x) - f(-x) = f(x) - h(x) - f(-x)$$

$$\Rightarrow 2h(x) = f(x) - f(-x)$$

$$\Rightarrow h(x) = \frac{1}{2}(f(x) - f(-x)) \in T.$$

So we can write $g(x)$, $h(x)$ only using $f(x)$ (1)

$\Rightarrow g, h$ are unique functions, with which we can write any f .

$$S \cap T = \{f\} \begin{cases} f \text{ odd} \Rightarrow f(-x) = -f(x) \\ f \text{ even} \Rightarrow f(-x) = f(x) \end{cases}$$

$$\begin{cases} f(-x) = -f(x) \\ f(-x) = f(x) \end{cases} \Rightarrow -f(x) = f(x) \Leftrightarrow f(x) = 0, (\forall) x \in \mathbb{R} \Rightarrow S \cap T = \{0\}.$$

$$\textcircled{c} \quad f, g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$h: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$a) \quad f(x, y) = (x+y, x-y)$$

$$b) \quad g(x, y) = (2x-y, 4x-2y)$$

$$c) \quad h(x, y, z) = (x-y, y-z, z-x)$$

$$f, g \in \text{End}_{\mathbb{R}}(\mathbb{R}^2)$$

$$h \in \text{End}_{\mathbb{R}}(\mathbb{R}^3)$$

$$f: V \rightarrow V'$$

$$K\text{-linear map: } \begin{cases} f(v_1 + v_2) = f(v_1) + f(v_2), & v_1, v_2 \in V & (1) \\ f(k \cdot v) = k \cdot f(v), & k \in K, v \in V & (2) \end{cases}$$

Endomorphism: K -linear map with $V = V'$.

a)

$$(1): \quad f(x, y) = (x+y, x-y)$$

$$f((x_1, y_1) + (x_2, y_2)) = f((x_1, y_1)) + f((x_2, y_2)) \quad (1)$$

$$\Rightarrow f(x_1 + x_2, y_1 + y_2) = (x_1 + x_2 + y_1 + y_2, x_1 + x_2 - y_1 - y_2)$$

$$= (\underline{x_1 + y_1} + \underline{x_2 + y_2}, \underline{x_1 - y_1} + \underline{x_2 - y_2})$$

$$= (x_1 + y_1, x_1 - y_1) + (x_2 + y_2, x_2 - y_2)$$

$$= f(x_1, y_1) + f(x_2, y_2)$$

$$(2): \quad f(k \cdot (x, y)) = f(kx, ky) = (kx + ky, kx - ky) = k(x + y, x - y) = k \cdot f(x, y)$$

$$(1), (2) \Rightarrow K\text{-linear map } \left. \begin{matrix} \mathbb{R}^2 = \mathbb{R}^2 \end{matrix} \right\} \Rightarrow f \in \text{End}_{\mathbb{R}}(\mathbb{R}^2)$$

$$\begin{aligned}
 b) \quad g((x_1, y_1) + (x_2, y_2)) &\stackrel{?}{=} g(x_1, y_1) + g(x_2, y_2) \\
 \Rightarrow g(x_1 + x_2, y_1 + y_2) &= (2(x_1 + x_2) - (y_1 + y_2), 4(x_1 + x_2) - 2(y_1 + y_2)) \\
 &= (2x_1 - y_1 + 2x_2 - y_2, 4x_1 - 2y_1 + 4x_2 - 2y_2) \\
 &= (2x_1 - y_1, 4x_1 - 2y_1) + (2x_2 - y_2, 4x_2 - 2y_2) \\
 &= g(x_1, y_1) + g(x_2, y_2) \\
 &= \mathbb{R}^2 = \mathbb{R}^2 \quad g(k(x, y)) = g(kx, ky) = (2kx - ky, 4kx - 2ky) \\
 &= k(2x - y, 4x - 2y) = k \cdot g(x, y) \\
 \Rightarrow \text{endomorphism } \checkmark
 \end{aligned}$$

$$\begin{aligned}
 c) \quad h((x_1, y_1, z_1) + (x_2, y_2, z_2)) &\stackrel{?}{=} h(x_1, y_1, z_1) + h(x_2, y_2, z_2) \\
 \Rightarrow h(x_1 + x_2, y_1 + y_2, z_1 + z_2) &= \\
 &= (x_1 + x_2 - y_1 - y_2, y_1 + y_2 - z_1 - z_2, z_1 + z_2 - x_1 - x_2) \\
 &= (\underbrace{x_1 - y_1} + \underbrace{x_2 - y_2}, \underbrace{y_1 - z_1} + \underbrace{y_2 - z_2}, \underbrace{z_1 - x_1} + \underbrace{z_2 - x_2}) \\
 &= (x_1 - y_1, y_1 - z_1, z_1 - x_1) + (x_2 - y_2, y_2 - z_2, z_2 - x_2) \\
 &= h(x_1, y_1, z_1) + h(x_2, y_2, z_2) \\
 \mathbb{R}^3 = \mathbb{R}^3 \Rightarrow \text{end } \checkmark \quad h(k(x, y, z)) &= \\
 &= h(kx, ky, kz) \\
 &= h(kx - ky, ky - kz, kz - kx) \\
 &= h(k(x - y), k(y - z), k(z - x)) \\
 &= k \cdot h(x - y, y - z, z - x) \\
 &= k \cdot h(x, y, z)
 \end{aligned}$$

* $\oplus$ Endomorphism?

$$a) \quad f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$f(x, y) = (ax + by, cx + dy)$$

$$\begin{cases}
 f(x+y) \stackrel{?}{=} f(x) + f(y) \\
 kf(x, y) = f(k \cdot (x, y)) \\
 \mathbb{R}^2 = \mathbb{R}^2 \checkmark
 \end{cases}$$

$$f(x+y) = (ax+by, cx+dy) = (ax$$

$$\begin{aligned}
 \bullet \quad f(x_1 + x_2, y_1 + y_2) &= (ax_1 + ax_2 + by_1 + by_2, cx_1 + cx_2 + dy_1 + dy_2) \\
 &= (ax_1 + by_1, cx_1 + dy_1) + (ax_2 + by_2, cx_2 + dy_2) \\
 &= f(x_1, y_1) + f(x_2, y_2)
 \end{aligned}$$

$$\begin{aligned}
 \bullet \quad f(k(x, y)) &= (kax + kby, kcx + kcy) = k(ax + by, cx + dy) = \\
 &= k f(x, y)
 \end{aligned}$$

$\Rightarrow f$ endo $\checkmark$

$$b) g: \mathbb{R}^2 \rightarrow \mathbb{R}^2, g(x, y) = (a+x, b+y)$$

$$a=b=0 \Rightarrow g(x, y) = (x, y) \rightarrow \text{endomorphism of } \mathbb{R}^2. \checkmark$$

$$(v) a, b \in \mathbb{R}^* \Rightarrow g(x_1+x_2, y_1+y_2) = (a+x_1+x_2, b+y_1+y_2) =$$

$$= (a+x_1, b+y_1) + (x_2, y_2)$$

$$= g(x_1, y_1) + (x_2, y_2)$$

$$\Rightarrow g \text{ is not an endom.}$$

$$\textcircled{8} f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$f(x, y) = (x \cdot \cos a - y \cdot \sin a, x \cdot \sin a + y \cdot \cos a)$$

$$f \in \text{End}_{\mathbb{R}}(\mathbb{R}^2) ?$$

$$(1) f(u+v) = f(u) + f(v)$$

$$(2) f(k \cdot u) = k \cdot f(u)$$

$$\bullet u = (x_1, y_1)$$

$$v = (x_2, y_2)$$

$$f(u+v) = f(x_1+x_2, y_1+y_2)$$

$$= ((x_1+x_2) \cos a - (y_1+y_2) \sin a, (x_1+x_2) \sin a + (y_1+y_2) \cos a)$$

$$= (x_1 \cos a - y_1 \sin a, x_1 \sin a + y_1 \cos a) + (x_2 \cos a - y_2 \sin a, x_2 \sin a + y_2 \cos a)$$

$$= f(x_1, y_1) + f(x_2, y_2) \quad (1)$$

$$\bullet f(k \cdot (x, y)) = f(kx, ky) =$$

$$= (kx \cos a - ky \sin a, kx \sin a + ky \cos a)$$

$$= k(x \cos a - y \sin a, x \sin a + y \cos a)$$

$$= k \cdot f(x, y) \quad (2)$$

$$\mathbb{R}^2 = \mathbb{R}^2 \xrightarrow[(2)]{(1)} f \text{ endom.}$$

$$f \rightarrow \text{multiplication with } A = \begin{pmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{pmatrix}$$

② $\text{Ker } f = ?$
 $\text{Im } f = ?$ $f(x, y) = (x+y, x-y)$

$f: V \rightarrow V'$ is a K -linear map.

$\text{Ker } f = \{v \in V \mid f(v) = 0\}$ $\rightarrow$ doesn't make sense for any function.

$\text{Im } f = \{f(v) \mid v \in V\}$. $\rightarrow$ makes sense for any function.

a) $f(x, y) = (x+y, x-y) = 0 \Rightarrow \begin{cases} x+y=0 \\ x-y=0 \end{cases}$
 $\xrightarrow{(+)} 2x=0 \Rightarrow x=0$
 $y=0$

$\Rightarrow \text{Ker}(f) = \{(0, 0)\} \Rightarrow f$ inj.

$\text{Im } f = \{(x+y, x-y) \mid x, y \in \mathbb{R}\} = \{(x-x) + (y, -y) \mid x, y \in \mathbb{R}\}$
 $= \{x(1, 1) + y(1, -1) \mid x, y \in \mathbb{R}\}$

$\Rightarrow \text{Im } f = \langle (1, 1), (1, -1) \rangle$.

$g(x, y) = (2x-y, 4x-2y)$ $\rightarrow a \in \mathbb{R}$

b) $g(x, y) = (0, 0) \Rightarrow \begin{cases} 2x-y=0 \\ 4x-2y=0 \end{cases} \Rightarrow 2x=y$

$x=a \in \mathbb{R} \Rightarrow y=2a \in \mathbb{R} \Rightarrow \text{Ker}(g) = \{(a, 2a) \mid a \in \mathbb{R}\}$
 $= \{a(1, 2) \mid a \in \mathbb{R}\} = \langle (1, 2) \rangle$.

$\text{Im } g = \{(2x-y, 4x-2y) \mid x, y \in \mathbb{R}\}$
 $= \{(2x, 4x) + (-y, -2y) \mid x, y \in \mathbb{R}\}$
 $= \{x(2, 4) + y(-1, -2) \mid x, y \in \mathbb{R}\}$

$\Rightarrow \text{Im } g = \langle (2, 4), (-1, -2) \rangle$.

$$c) h(x, y, z) = (x-y, y-z, z-x)$$

$$\ker(h) = \{(x, y, z) \mid (x-y, y-z, z-x) = (0, 0, 0)\}$$

$$\Rightarrow \begin{cases} x-y=0 \\ y-z=0 \\ z-x=0 \end{cases}$$

$$(1)+(3) \Rightarrow \begin{cases} z-y=0 \\ y-z=0 \end{cases} \Rightarrow x=y=z$$

$$\begin{aligned} \Rightarrow \ker(h) &= \{(x, x, x) \mid x \in \mathbb{R}\} \\ &= \{x(1, 1, 1) \mid x \in \mathbb{R}\} \\ &= \langle (1, 1, 1) \rangle. \end{aligned}$$

$$\begin{aligned} \operatorname{Im} h &= \{(x-y, y-z, z-x) \mid \dots\} \\ &= \{(x, 0, x) + (-y, y, 0) + (0, -z, z) \mid \dots\} \\ &= \{x(1, 0, 1) + y(-1, 1, 0) + z(0, -1, 1) \mid \dots\} \\ &= \langle (1, 0, 1), (-1, 1, 0), (0, -1, 1) \rangle \end{aligned}$$

V vector space, let $v_1, v_2, \dots, v_m \in V$.

$\Rightarrow \langle v_1, v_2, \dots, v_m \rangle$ generated subspace if

$$(\forall) v \in V, \exists \underbrace{\alpha_1, \alpha_2, \dots, \alpha_m}_{\in K} \text{ s.t. } \alpha_1 v_1 + \dots + \alpha_m v_m = v$$

$$V = A \oplus B \text{ "direct sum of A and B" if } \begin{cases} V = A + B \\ A \cap B = \{0\} \end{cases}$$

$a+b$

$$\text{or } (\forall) v \in V, \exists! a \in A \text{ s.t. } v = a + b. \\ \exists! b \in B \quad (v = \alpha a + \beta b)$$

$f: A \rightarrow B$ linear map
(homomorphism of vector spaces)

$$\text{if: } \begin{cases} (\forall) v_1, v_2 \in A: f(v_1 + v_2) = f(v_1) + f(v_2) \\ (\forall) v \in A, (\forall) \alpha \in K: f(\alpha \cdot v) = \alpha f(v) \end{cases}$$

$$f \in {}_K \text{End}(A) \text{ if: } \begin{cases} A = B \\ f \text{ homomorphism.} \end{cases}$$

(endomorphism)

$$\text{Ker } f = \{x \in A \mid f(x) = 0\}.$$

$$\text{Im } f = \{y \in B \mid f(x) = y\}$$

① a) $\langle 1, x, x^2 \rangle = \{ \alpha_1 \cdot 1 + \alpha_2 \cdot x + \alpha_3 \cdot x^2 \mid \alpha_1, \alpha_2, \alpha_3 \in K \}$
generated subspace given by vectors $1, x, x^2$.
 $= \{ f \in K[x] \mid \text{degree}(f) \leq 2 \}.$

b) $\langle \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \rangle = M_2(K) = \left\{ \begin{pmatrix} \alpha_1 & \alpha_2 \\ \alpha_3 & \alpha_4 \end{pmatrix} \right\}$

$\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in K$

c) $\langle (1, 0), (0, 1) \rangle = K^2$

d) $\langle (1, 0), (2, 0) \rangle = K = \{ (x, 0) \in K^2 \mid x \in K \}.$

K^4 :
 $(0, 1, 1, 1)$

$\mathbb{C} = \langle 1, i \rangle = \{ \alpha_1 \cdot 1 + \alpha_2 \cdot i \dots \}, \alpha_1, \alpha_2 \in \mathbb{R}$

$$\textcircled{2} \quad A = \{(x, y, z) \in \mathbb{R}^3 \mid x=0\}$$

$$B = \{(x, y, z) \in \mathbb{R}^3 \mid x+y+z=0\}$$

$$C = \{(x, y, z) \in \mathbb{R}^3 \mid x=y=z\}.$$

as generated subspaces?

$$\begin{aligned} \text{a) } B &= \{(x, y, z) \in \mathbb{R}^3 \mid x = -y - z\} \\ &= \{(-y - z, y, z) \in \mathbb{R}^3 \mid y, z \in \mathbb{R}\} \\ &= \{(-y, y, 0) + (-z, 0, z) \mid y, z \in \mathbb{R}\}. \\ &= \{y(-1, 1, 0) + z(-1, 0, 1) \mid y, z \in \mathbb{R}\}. \\ &= \langle (-1, 1, 0), (-1, 0, 1) \rangle. \end{aligned}$$

$y, z \rightarrow$ treat them as scalars.

$$\begin{aligned} \text{a) } A &= \{(x, y, z) \in \mathbb{R}^3 \mid x=0\} \\ &= \{(0, y, z) \in \mathbb{R}^3 \mid y, z \in \mathbb{R}\}. \\ &= \{y(0, 1, 0) + z(0, 0, 1) \mid y, z \in \mathbb{R}\}. \\ &= \langle (0, 1, 0), (0, 0, 1) \rangle. \end{aligned}$$

$$\begin{aligned} \text{c) } C &= \{(x, x, x) \in \mathbb{R}^3 \mid x \in \mathbb{R}\}. \\ &= \{x(1, 1, 1) \in \mathbb{R}^3 \mid x \in \mathbb{R}\} \\ &= \langle (1, 1, 1) \rangle \end{aligned}$$

$$\begin{aligned} \mathbb{R}^3 &= \{(x, y, z) \mid x, y, z \in \mathbb{R}\} \\ &= \{(x, 0, 0) + (0, y, 0) + (0, 0, z) \mid x, y, z \in \mathbb{R}\}. \\ &= \{x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1) \mid x, y, z \in \mathbb{R}\}. \\ &= \langle (1, 0, 0), (0, 1, 0), (0, 0, 1) \rangle. \end{aligned}$$

base of the $\mathbb{R}^3$ vector space

⑤ $\mathbb{R}^3$. $a = (-2, 1, 3)$, $b = (3, -2, -1)$, $c = (1, -1, 2)$, $d = (-5, 3, 4)$
 $e = (-9, 5, 10)$

-6

$$\langle a, b \rangle = \langle c, d, e \rangle$$

(*) $v \in \mathbb{R}^3$

$$\langle a, b \rangle \Rightarrow v = \alpha_1 \cdot a + \alpha_2 \cdot b$$

$$\langle c, d, e \rangle \Rightarrow v = \beta_1 \cdot c + \beta_2 \cdot d + \beta_3 \cdot e$$

But : if
$$\begin{cases} a = \gamma_1 \cdot c + \gamma_2 \cdot d + \gamma_3 \cdot e \\ b = \delta_1 \cdot c + \delta_2 \cdot d + \delta_3 \cdot e \end{cases}$$

$$c + e = (1, -1, 2) + (-9, 5, 10) = (-8, 4, 12) = 4a$$

$$\Rightarrow a = \frac{1}{4} \cdot c + 0 \cdot d + \frac{1}{4} \cdot e$$

$$b = \delta_1(1, -1, 2) + \delta_2(-5, 3, 4) + \delta_3(-9, 5, 10)$$

$$\Rightarrow \begin{cases} 3 = \delta_1 - 5\delta_2 - 9\delta_3 \\ -2 = -\delta_1 + 3\delta_2 + 5\delta_3 \\ -1 = 2\delta_1 + 4\delta_2 + 10\delta_3 \end{cases}$$

$$\Rightarrow A = \begin{pmatrix} 1 & -5 & -9 \\ -1 & 3 & 5 \\ 2 & 4 & 10 \end{pmatrix}$$

$$\det A = \begin{vmatrix} 1 & -5 & -9 \\ -1 & 3 & 5 \\ 2 & 4 & 10 \end{vmatrix} = \dots = 0$$

$$d_2 = \begin{vmatrix} 1 & -5 \\ -1 & 3 \end{vmatrix} = -2 \neq 0 \Rightarrow \text{rank } A = 2.$$

$$\Delta c = \begin{vmatrix} -1 & -5 & 3 \\ -1 & 3 & -2 \\ 2 & 4 & -1 \end{vmatrix} = 0.$$

$$\Rightarrow \text{Take } \delta_3 = c' \in \mathbb{R}.$$

$$\delta_1 - 2\delta_2 = 3 + 9c'$$

$$-\delta_1 + 3\delta_2 = -2 - 5c'$$

$$c = a + b$$

$$d = a - b$$

$$e = 3a - b$$

$$f_2 = -\frac{1}{2} - 2c'$$

$$d_1 = -c' + \frac{1}{2}$$

$$\Rightarrow b = (-c' + \frac{1}{2}) \cdot c + (-\frac{1}{2} - 2c') \cdot d + c' \cdot e, (\forall) c' \in \mathbb{R}.$$

$$(4) S = \{(x, y, z) \in \mathbb{R}^3 \mid x+y+z=0\} = \langle \underbrace{(-1, 1, 0)}_{s_1}, \underbrace{(-1, 0, 1)}_{s_2} \rangle$$

$$T = \{(x, y, z) \in \mathbb{R}^3 \mid x=y=z\} = \langle \underbrace{(1, 1, 1)}_t \rangle.$$

S, T subspaces? (See seminar 4).

$$\mathbb{R}^3 = S \oplus T.$$

$$(\forall) v \in \mathbb{R}^3, \exists! s \in S, \exists! t \in T \text{ s.t. } v = s + t = \alpha \cdot s + \beta \cdot t$$

$$v = \alpha(s_1 + s_2) + \beta \cdot t = \alpha_1 \cdot s_1 + \alpha_2 \cdot s_2$$

$$\rightarrow v = \alpha \cdot (\alpha_1 \cdot s_1 + \alpha_2 \cdot s_2) + \beta \cdot t$$

$$v = \underbrace{\alpha \cdot \alpha_1}_{\alpha_1} \cdot s_1 + \underbrace{\alpha \cdot \alpha_2}_{\alpha_2} \cdot s_2 + \beta \cdot t$$

$$v = \alpha_1 \cdot s_1 + \alpha_2 \cdot s_2 + \beta \cdot t$$

$$v \in \mathbb{R}^3 \Rightarrow v = (v_1, v_2, v_3)$$

$$\Rightarrow (v_1, v_2, v_3) = (-\alpha_1 - \alpha_2 + \beta, \alpha_1 + \beta, \alpha_2 + \beta)$$

$$\Rightarrow \begin{cases} -\alpha_1 - \alpha_2 + \beta = v_1 & (1) \\ +\alpha_1 + \beta = v_2 \Rightarrow \alpha_1 = v_2 - \beta \\ +\alpha_2 + \beta = v_3 \Rightarrow \alpha_2 = -\beta + v_3 \end{cases} \Rightarrow (1):$$

$$\Rightarrow -v_2 + \cancel{\beta} + \cancel{\beta} + v_3 + \beta = v_1.$$

$$\Rightarrow 3\beta = v_1 + v_2 + v_3 \Rightarrow \beta = \frac{v_1 + v_2 + v_3}{3}, (\forall) v_1, v_2, v_3 \in \mathbb{R}.$$

$$\Rightarrow \alpha_1 = v_2 - \beta = v_2 - \frac{v_1 + v_2 + v_3}{3} = \frac{2v_2 - v_1 - v_3}{3}, (\forall) v_1, v_2, v_3 \in \mathbb{R}$$

$$\Rightarrow \alpha_2 = -\beta + v_3 = \dots \Rightarrow \alpha_2 = \frac{2v_3 - v_1 - v_2}{3}, (\forall) v_1, v_2, v_3 \in \mathbb{R}$$

Ex: $v = (1, 0, -1)$

$$\alpha_1 = \frac{2 \cdot 0 - 1 + 1}{3} = 0$$

$$\alpha_2 = -1$$

$$\beta = 0. \quad \left. \begin{array}{l} \Rightarrow v = (0 \cdot s_1 - 1 \cdot s_2 + 0 \cdot t) \\ \Rightarrow v = -s_2. \end{array} \right\}$$

5) $f(x,y) = (x+y, x-y)$

$f: g \rightarrow$

$$\begin{cases} f((x_1, y_1) + (x_2, y_2)) \stackrel{?}{=} f(x_1, y_1) + f(x_2, y_2) \\ f(k(x, y)) \stackrel{?}{=} k \cdot f(x, y) \end{cases}$$

$$\text{Ker}(f) = \{(0, 0)\}.$$

$$\text{Im}f = \langle (\quad), (\quad) \rangle.$$

$$g(x, y) = (2x - y, 4x - 2y)$$

$$v_1 = (x_1, y_1)$$

$$v_2 = (x_2, y_2)$$

$$(1): g(v_1 + v_2) \stackrel{?}{=} g(v_1) + g(v_2)$$

ist:

$$\Leftrightarrow (2x_1 - y_1 + 2x_2 - y_2, 4x_1 - 2y_1 + 4x_2 - 2y_2) \stackrel{?}{=}$$

$$(2x_1 - y_1, 4x_1 - 2y_1) + (2x_2 - y_2, 4x_2 - 2y_2)$$

$$\Leftrightarrow (2x_1 - y_1 + 2x_2 - y_2, 4x_1 - 2y_1 + 4x_2 - 2y_2) = g(v_1) + g(v_2). \text{ true}$$

$$(2): g(k \cdot v) = k \cdot g(v)$$

$$\Rightarrow g(k(x_1, y_1)) = g(kx_1, ky_1) = (2kx_1 - ky_1, 4kx_1 - 2ky_1) = (k(2x_1 - y_1), k(4x_1 - 2y_1)). \text{ true}$$

$$\text{Ker}(f) = \{(x, y) \in \mathbb{R}_2 \mid g(x, y) = (0, 0)\}.$$

$$\Rightarrow (2x - y, 4x - 2y) = (0, 0)$$

$$\Rightarrow \begin{cases} 2x - y = 0 & \Rightarrow y = 2x \\ 4x - 2y = 0 & | :2 \Rightarrow y = 2x \end{cases}$$

$$\Rightarrow \{(x, 2x) \mid x \in \mathbb{R}\} = \{x(1, 2) \mid x \in \mathbb{R}\} = \langle (1, 2) \rangle.$$

$$\text{Im}f = \{(2x + y, 4x - 2y) \mid f(x, y) = (2x - y, 4x - 2y)\}$$

$$= \{(2x, 4x) + (-y, -2y) \mid x, y \in \mathbb{R}\}$$

$$= \{2x(1, 2) + (-y)(1, 2) \mid x, y \in \mathbb{R}\} = \langle (1, 2), (-1, -2) \rangle$$